



**Faculty of Science and Arts
Department of Chemistry
Biochemistry Lab (CHEM 266)**

Experiment 1

AMINO ACIDS TITRATION

Pre-Lab Questions:

- 1) If the equilibrium constant, K_a , for the ionization of the carboxylic acid group is 1×10^{-3} , what is the pK_a ?
- 2) If in a solution of alanine, the number of negatively charged carboxylate groups (COO^-) is 1×10^{20} at the isoelectric point, what is the number of positive charged amino groups, NH_3^+ ?

Precautions

Waste disposal: Please dispose the liquid wastes in a large beaker, the lab technician needs to adjust the pH before pouring the liquid waste down the drain.

Objectives:

1. To perform one of the most common method of analysis, acid-base titration.
2. To draw the titration curve for an amino acid.
3. To use this curve to estimate the pK_a values of the ionizable groups of the amino acid.
4. To understand the acid-base behavior of an amino acid.

Introduction

Amino acids are the building blocks of proteins. They contain an amino group and a carboxyl group both of which are bonded to a central chiral carbon atom called α -carbon. The α -carbon is also bonded to hydrogen and side chain group, which is represented by the letter **R**. The R group determines the identity and properties of the particular amino acid (figure 1). Among all the possible amino acids, only 20 are usually found in proteins. The one-letter and three letters abbreviations of the standard amino acids are presented in Table 1.

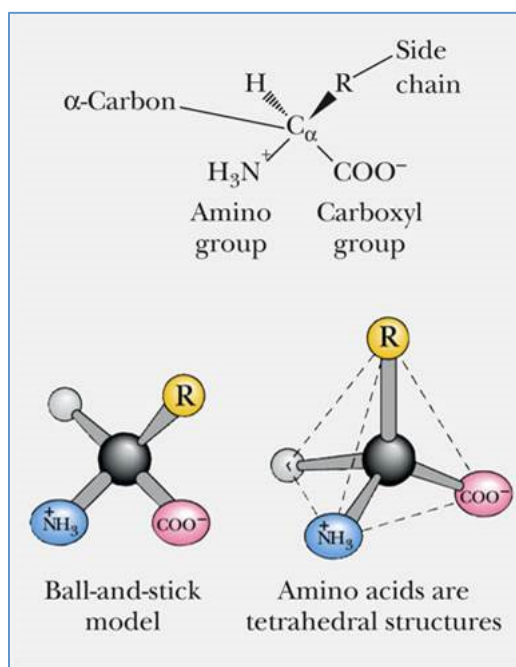


Figure 1. The general formula of amino acids, showing the ionic form that predominates at pH 7.4.

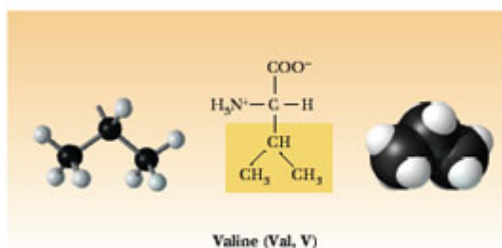
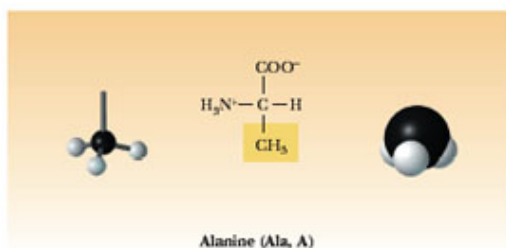
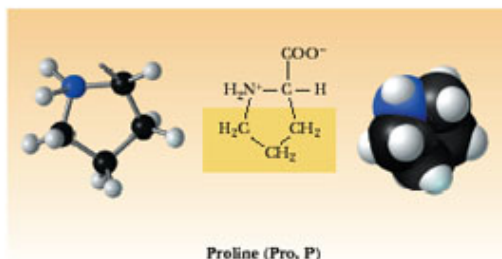
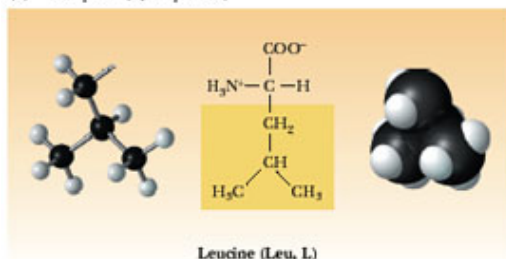
Table 1. Names and abbreviations of the common amino acids

Names and Abbreviations of the Common Amino Acids		
Amino Acid	Three-Letter Abbreviation	One-Letter Abbreviation
Alanine	Ala	A
Arginine	Arg	R
Asparagine	Asn	N
Aspartic acid	Asp	D
Cysteine	Cys	C
Glutamic acid	Glu	E
Glutamine	Gln	Q
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Leucine	Leu	L
Lysine	Lys	K
Methionine	Met	M
Phenylalanine	Phe	F
Proline	Pro	P
Serine	Ser	S
Threonine	Thr	T
Tryptophan	Trp	W
Tyrosine	Tyr	Y
Valine	Val	V

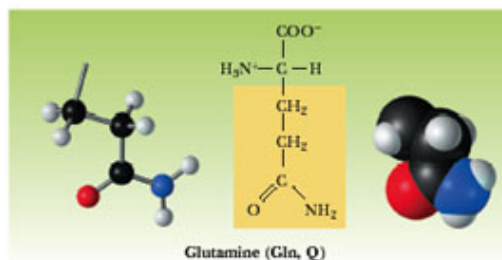
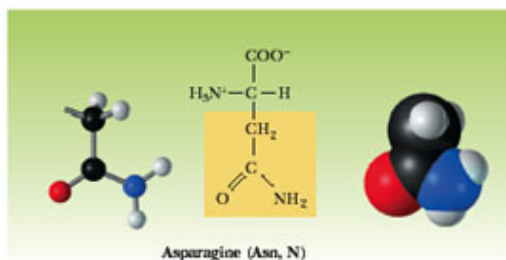
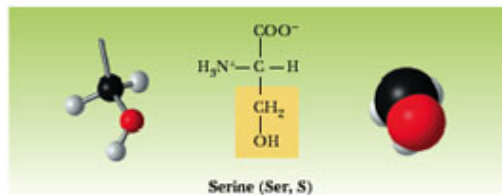
The 20 amino acids are classified based on the R group. Amino acids might be polar (hydrophilic) or non-polar (hydrophobic) depending on the polarity of R group, or they might be acidic, basic, and neutral depending on the titration properties of amino acid R group. For each amino acid, the ball-and-stick model (*left*) and the space-filling model (*right*) show only the side chain as presented in Table 2.

Table 2. The chemical structure of the 20 standard amino acids

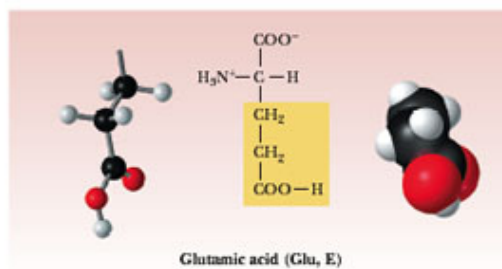
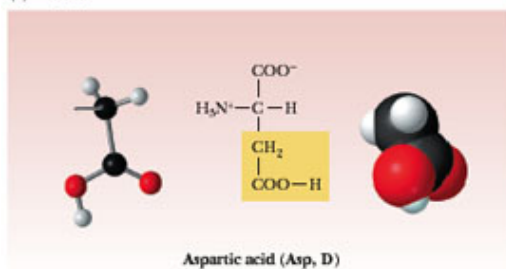
(a) Non-polar (hydrophobic)

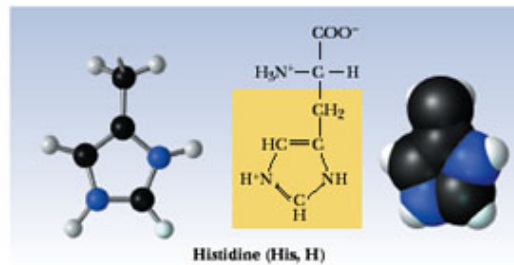
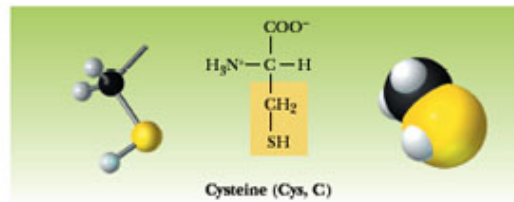
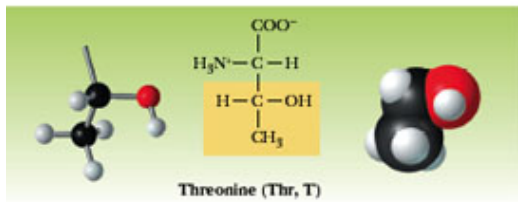
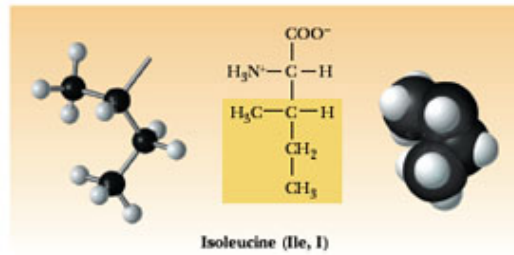
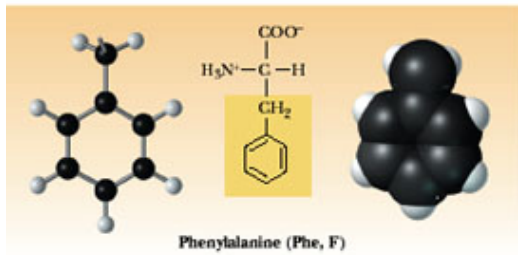
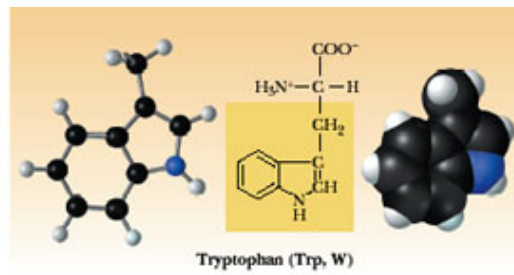
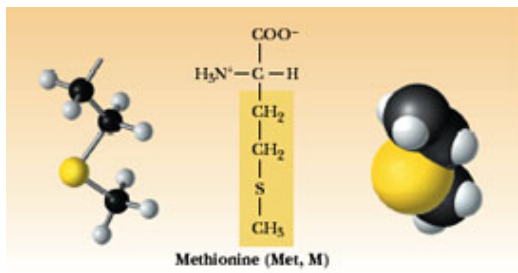


(b) Polar, uncharged

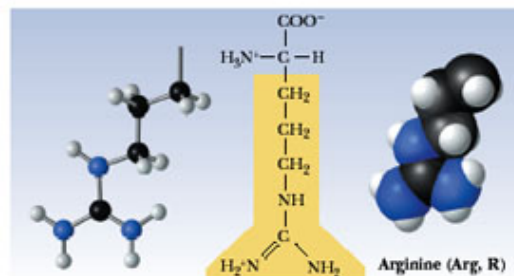
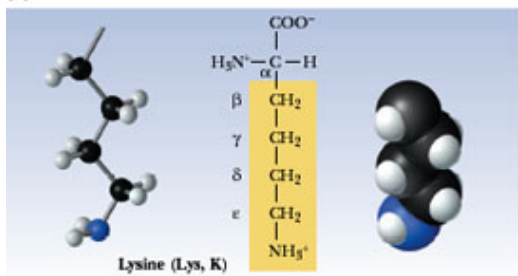


(c) Acidic





(d) Basic



Ionization of Amino Acids

Amino acids without charged groups on side chains (neutral amino acids) exist in neutral solution as isoelectric zwitterion molecule (figure 2). A zwitterion molecule has equal positive and negative charges. For example, at very low pH, alanine has a protonated (uncharged) carboxyl group and a protonated (positively charged) amino group. Under these conditions, the net charge of alanine is positive 1. As base is added the pH in the solution starts to increase; at a pH above 2.3 (pK_a of the carboxyl group) carboxyl group loses its proton to become a negatively charged carboxylate group (figure 2). As the pH increases still further with addition of more bases, the pH in the solution becomes above the pK_a of the amino group, which is 9.69 and it loses its proton, and the alanine molecule now has a negative charge of 1.

Notice that there is an inverse relationship between pK_a and acidity of the groups on the amino acid. The first group to lose its proton during the titration process is the group with the lowest pK_a value. Notice also that amino acids are considered amphoteric molecules it can react as an acid when base is added and it will react as base when acid is added and this can be demonstrated in image below by stating the equilibrium of the reaction in both directions.

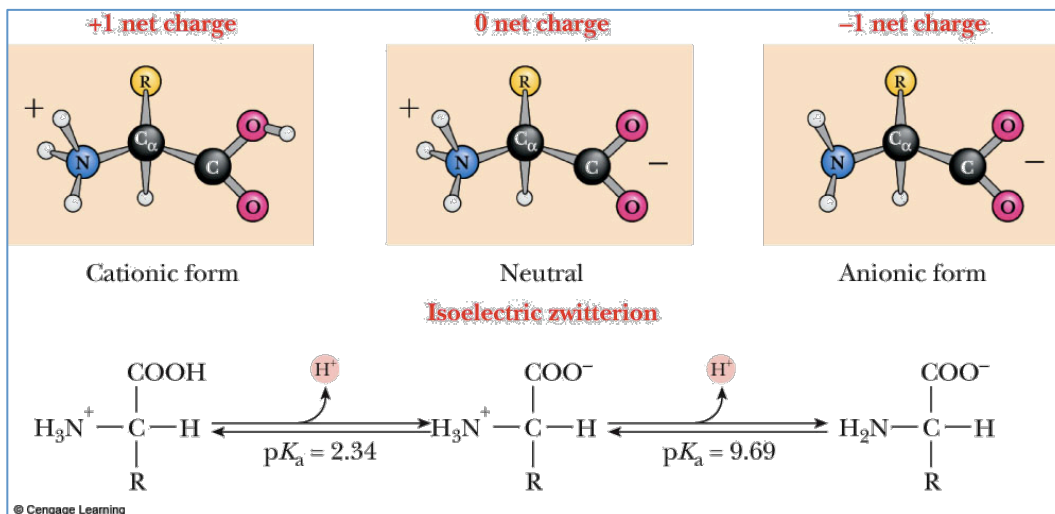


Figure 2. Ionization of alanine amino acid.

Table 3. pK_a values of standard amino acids.

pK_a Values of Common Amino Acids			
Acid	α-COOH	α-NH₃⁺	RH or RH⁺
Gly	2.34	9.60	
Ala	2.34	9.69	
Val	2.32	9.62	
Leu	2.36	9.68	
Ile	2.36	9.68	
Ser	2.21	9.15	
Thr	2.63	10.43	
Met	2.28	9.21	
Phe	1.83	9.13	
Trp	2.38	9.39	
Asn	2.02	8.80	
Gln	2.17	9.13	
Pro	1.99	10.6	
Asp	2.09	9.82	3.86*
Glu	2.19	9.67	4.25*
His	1.82	9.17	6.0*
Cys	1.71	10.78	8.33*
Tyr	2.20	9.11	10.07
Lys	2.18	8.95	10.53
Arg	2.17	9.04	12.48

* For these amino acids, the R group ionization occurs before the α -NH₃⁺ ionization.

Titration of Amino Acids

A titration is an analytical method in which measured amounts of base is added to a measured amount of acid. It is convenient and straightforward to follow the course of the reaction with pH meter. The point in the titration at which the acid is exactly neutralized with the base is called the equivalence point. When an amino acid is titrated, the titration curve represents the reaction of each functional group with the hydroxide ion. The amino acids have different pK_a values, which rise to the possibility of having different charges at a given pH. The pH at which a molecule has no net charge is called isoelectric pH, or isoelectric point (pI), isoelectric pH is the point at which the molecule has the sum of positive charges equals the sum of negative charges, and it can also be defined as the pH at which the amino acid is electrically neutral and will not migrate in

an electric field. The pI of an amino acid can be calculated by the following equation:

$$pI = \frac{pK_{a1} + pK_{a2}}{2}$$

Most amino acids have only two pK_a values the first is corresponding to the carboxylic group and the second to the amino group. For the acidic and basic amino acids, we must be sure to average the correct pK_a values. The pI of acidic amino acids is the average of pK_{a1} and pK_{a2} , while the pI for basic amino acids is the average of pK_{a2} and pK_{a3} . Around the inflection points is what is called the buffer region, which can be defined as the region where there is a resistance or minimal changes on the pH of the solution when strong base/acid is added. It equals pK_a value for the group +1 and -1 ($pH=pK_a \pm 1$). Amino acids contain a weakly acidic carboxylic groups and weak basic amino groups. The relationship between the concentration of weak acid and its conjugate base is described by Henderson-Hasselbalch equation:

$$pH = pK_a + \log \frac{[\text{Conjugate base}]}{[\text{Weak acid}]}$$

When $[\text{weak acid}] = [\text{conjugate base}]$, the $pH = pK_a$ and this point is called the inflection point which is the point when 0.5 mole of base has been added to 0.5 mole of acid present. At very low pH the amino group has a positive charge and the carboxyl group has zero charge. As the base is added and the pH increases, the carboxyl group loses a proton to become a carboxylate $-\text{COO}^-$. With adding more base the amino group or the R group (based on their pK_a values) loses its proton to the medium.

1. Titration curve of neutral amino acid

Alanine is an example of amino acid, which has 2 acidic protons (diprotic acid).

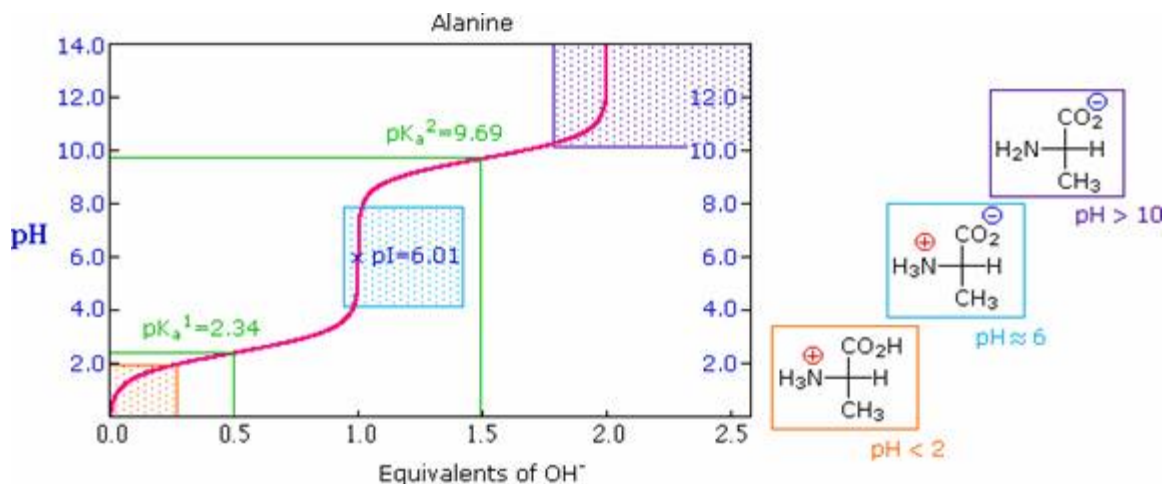


Figure 3. The titration curve of alanine <http://chemistryjee.blogspot.com/2015/06/>

2. Titration curve of an acidic amino acid

Aspartic acid is an example of an acidic amino acid, which has 3 titrable protons (triprotic acids).

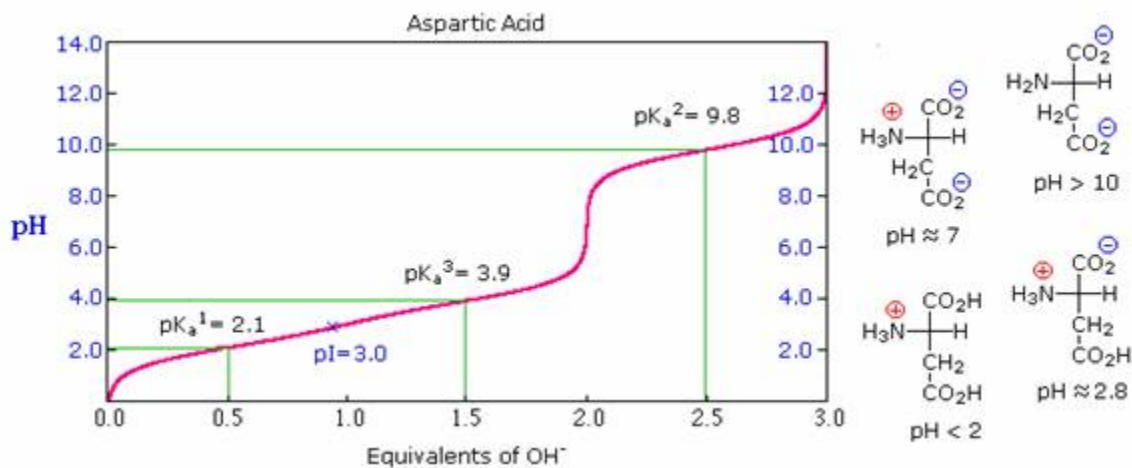


Figure 4. The titration curve of aspartic acid <http://chemistryjee.blogspot.com/2015/06/>

3. Titration curve of a basic amino acid

Arginine is an example of basic amino acid, which has 3 titratable protons (triprotic acid)

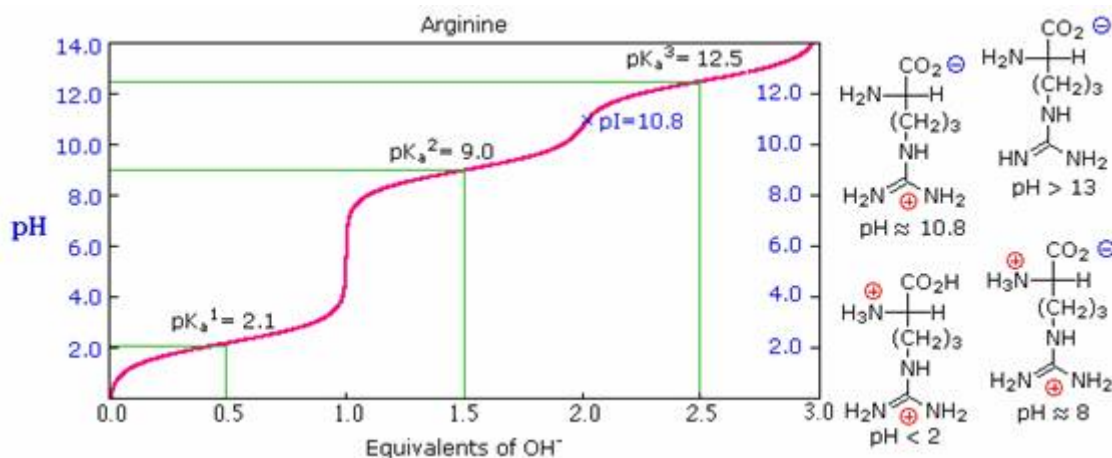


Figure 5. The titration curve of arginine <http://chemistryjee.blogspot.com/2015/06/>

Experimental procedure

Materials needed

1. Hydrochloric acid (100 mmol/L)
2. Sodium hydroxide (100 mmol/L)
3. Amino acids: (100 mmol/L alanine, 100 mmol/L lysine, 50 mmol/L aspartic acid)

Procedure

1. Place 10 mL of the amino acid solution into a 100 mL beaker (figure 6).
2. Measure the pH of the solution using the pH meter.
3. Add the 100 mmol/L HCL from a burette in small amounts (2-3 mL) at first and determine the pH after each addition until the pH falls to about 1.3.
4. Titrate a further amino acid solution with 100 mmol/L NaOH solution to pH 12.5.
5. Plot your data points on a graph, pH on Y-axis and volumes of NaOH added on X-axis
6. Determine the pK_a values and isoelectric point from your curve.

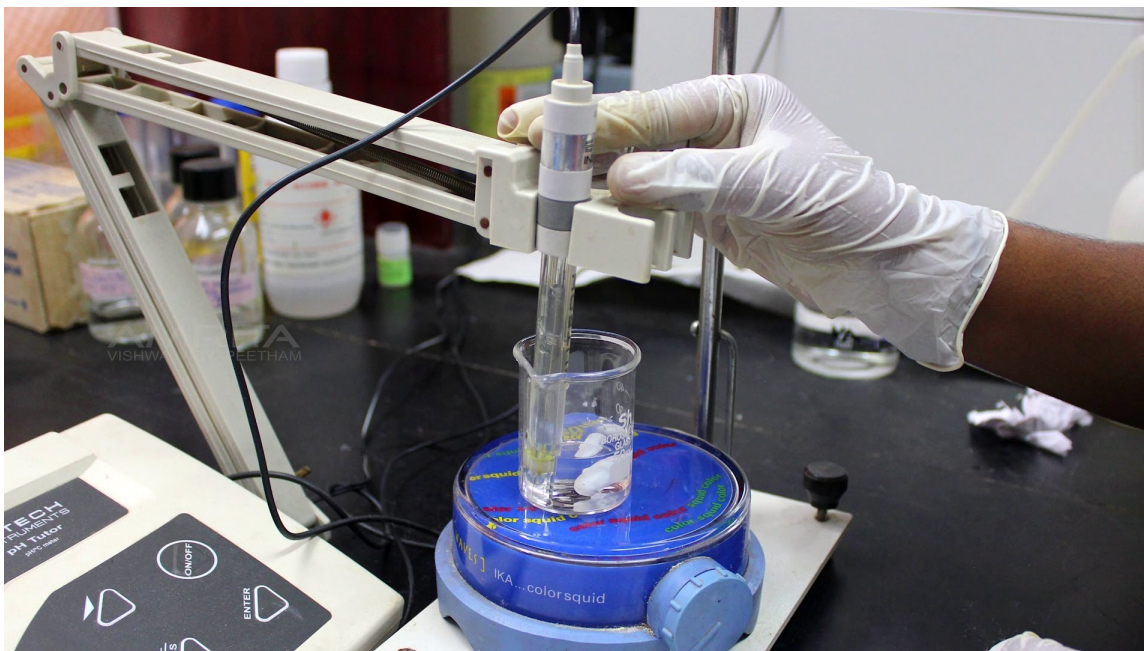


Figure 6. Amino acid titration. <http://oxy-tops.org/programs/chemistry-program/equilibrium/>

Data Analysis

1. Write the report of this experiment in your own words.
2. Prepare a graph of your results, plotting moL of NaOH added versus pH. Plot each of the two titrations (not the dry run) on a separate sheet of graph paper. Actually, I recommend using Excel for this. Remember to label the graph axis and provide enough tick marks along each axis to be useful; also, give the graph an appropriate title. It is easy to forget these steps when using software; write the information in by hand, if necessary.
3. Graphically determine the values of pKa's for your amino acid from your titration graphs. Each graph should show the location of the pKa's and the numerical value you determined for each. Remember, pKa's are at half- equivalence points.
4. The determining of the pI graphically is usually not recommended. Instead, it is determined by averaging the pKa's of the two adjoining buffer regions, which can be determined with reasonable accuracy from a good graph, since they lie in the relatively flat buffer regions. Show the location of the isoelectric point, with the pI you calculated for it on each graph.
5. Identify your amino acid

References:

1. Campbell, M.; Farrell, S. *Biochemistry*, 7th ed.; Brook/Cole, CA, 2009.



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Titration Curves of Amino Acids

Report Author:

Lab Partner:

Date Due:

Date Turned In:

Unknown #:

Unknown Identification:

Purpose:**Data and calculations:**

Note: On a graph paper plot pH versus volume (mL) NaOH.

From the curves

Amino acid	pKa ₁	pKa ₂	pKa ₃	pI

Conclusion:

Errors and problems you encountered and their possible remedies